



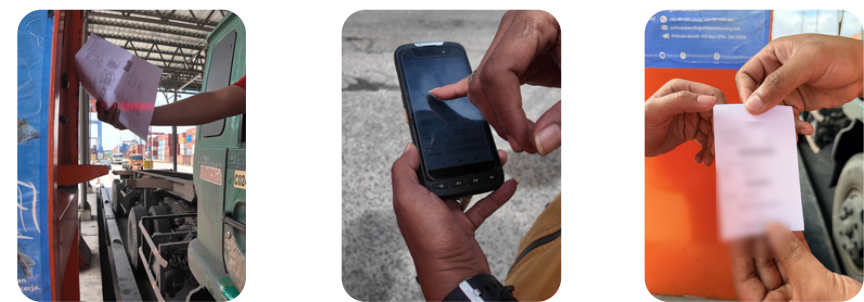
SYSTEM ARCHITECTURE DESIGN FOR AUTOMATED CONTAINER INSPECTION IN PORTS

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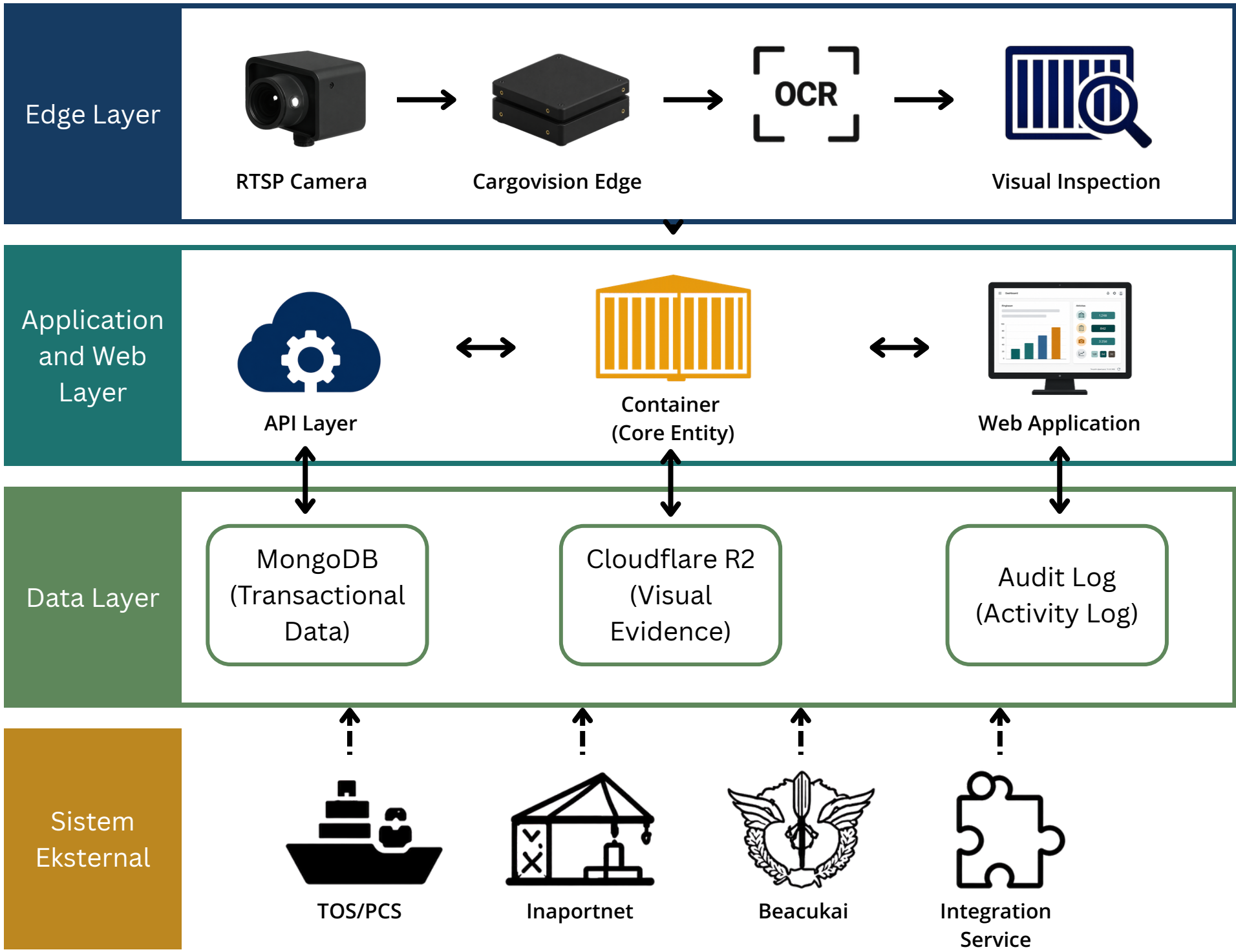
1 EXISTING CONDITIONS

Container inspection information is fragmented across multiple media, disconnected from inspection history, and heavily dependent on manual recording.

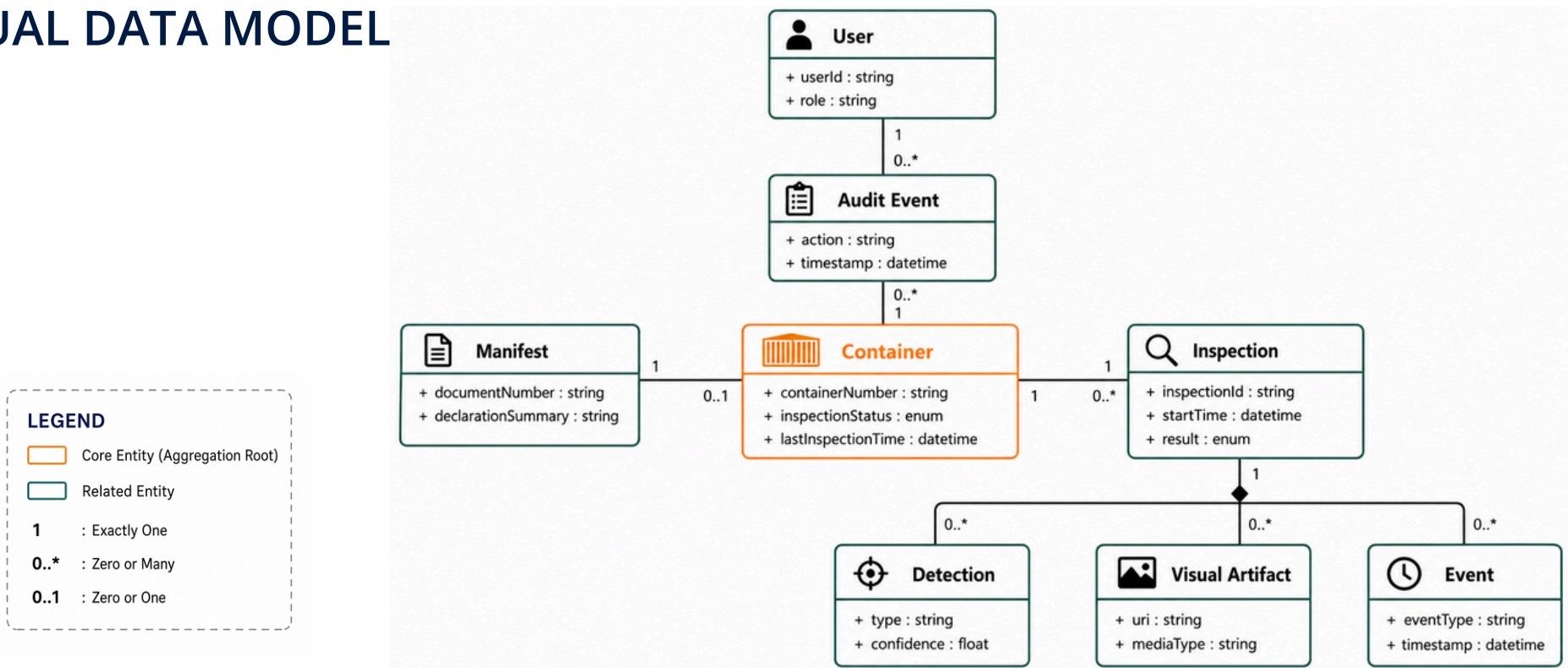


- Paper-Based Documentation
 - Manual Data Entry
 - Physical Container Inspection
- Inspection information is scattered across multiple media and systems
 - Visual evidence is not linked to inspection records
 - Container history is difficult to trace and audit
 - Inspection activities still rely on manual recording

2 CONTAINER-CENTRIC CARGOVISION ARCHITECTURE



3 CONCEPTUAL DATA MODEL



4 CAPACITY EVALUATION SUMMARY (SCENARIO-BASED)



Capacity estimation based on inspection-cycle scenarios, lane configuration, and operational utilization.

5 KESIMPULAN

The proposed architecture positions the **container** as the **central inspection-history entity**, separates responsibilities across edge, API, storage, and web components, and supports scenario-based capacity evaluation.

